EV Forecast driven Load Shifting with Economic Model Predictive Control

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Research Overview

The rapid growth of Electric Vehicles (EVs) poses challenges on both electricity demand and supply balance and the grid stability. This work proposes an innovative decentralized approach of optimizing EV charging profiles with a Mixed-Integer Linear Programming (MILP) technique, namely Model Predictive Control (MPC) driven by forecasting on EV charging variables. A concise framework is developed which updates EV energy demand (ED) in real-time, dynamically adjusting charging schedules to shift load and minimize peak demand. Simulation results show obvious peak loads reduction and monetary cost reduction, offering a possible solution for future EV charging aggregator.

Methodology

In this work 6 cases are consider for comparison:
V0G: No optimization but full power charging upon arrival of each EV
No forecast: Optimization with arrived EV sessions only
Perfect forecast: Assume all time and energy are known prior to test
Persistence forecast: Use last persistence day's real sessions, classified in weekdays, weekends and holidays
Statistic forecast: Use last several days' sessions, keeping time unchanged and linearly scaling ED
LSTM: Build a neural network model to forecast ED time series

Algorithm 1 Model Predictive Control Workflow

Require: Historical training sessions, variables setting, forecast method selecting
1: **Initialize** test data loading
2: **for** day in test days **do**
3: 00:00 Perform daily forecast on N_{EV}, AT, DT, and ED
4: **for** 00:15 to 23:45 **do**
5: Perform real-time forecast on N_{EV}, AT, DT, and ED
6: Combine arrived EV session with new forecast
7: Solve the optimization problem:
$$\min_{L,P,E} J(k,L)$$
8: Implement the current load variables $L(k)$, $P(k)$ and $E(k)$
9: **end for**
10: **end for**
11: Calculating the total electricity cost and plotting the load time series

Case Study

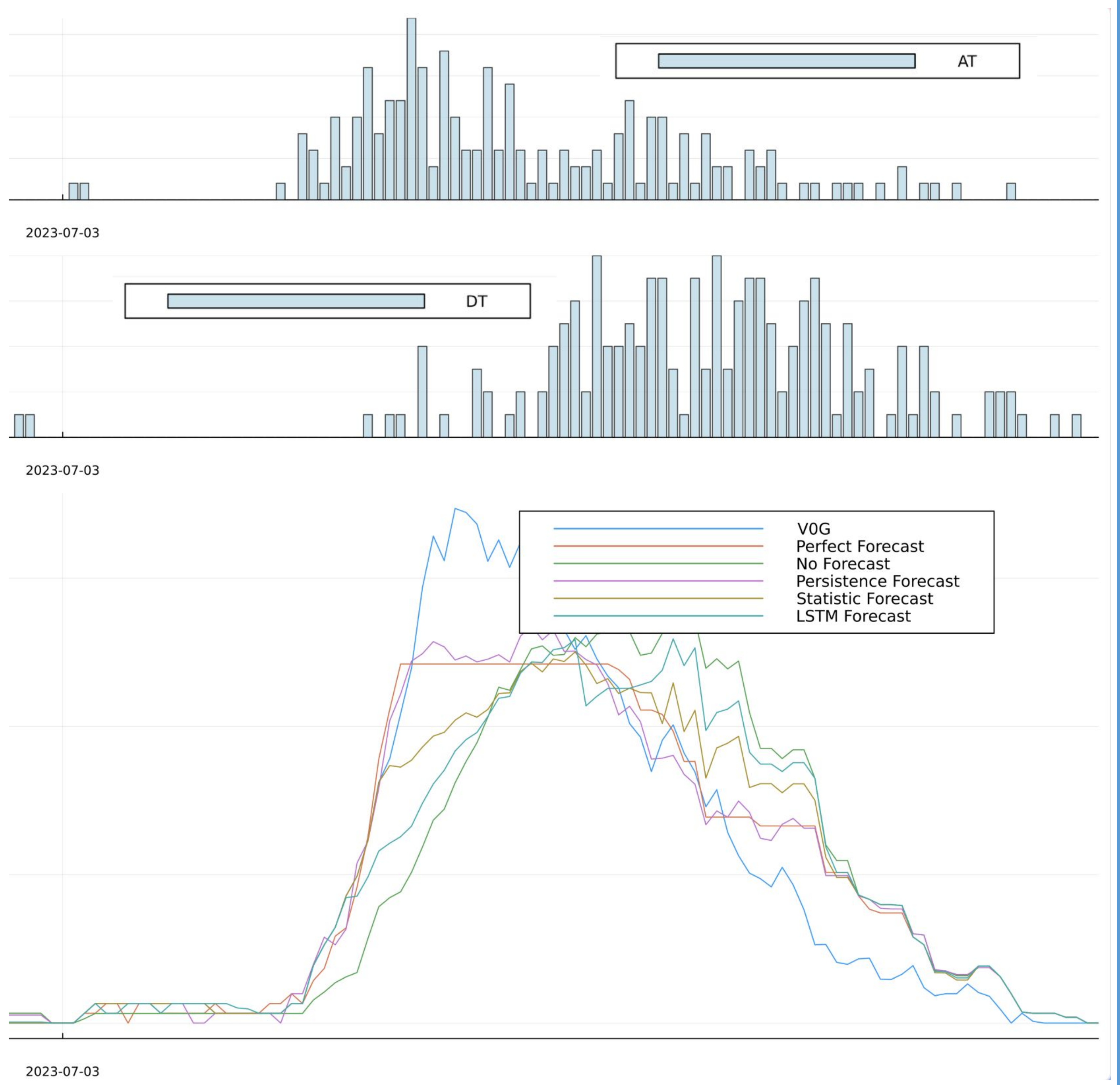
Simulation uses 3 days of 2023 summer, with 2 weekends and 1 weekday. V0G in blue shows the biggest load fluctuation, with the perfect forecast in orange being the most flat during peak daytime. Intuitively, methods are better if they are closer to the perfect forecast line. 3 forecast driven load lines have similar trends as the perfect one, while the no forecast line in green has a lagging peak load, with a higher peak demand. Among all non-perfect methods, the persistence forecast is the best with the flattest load profile and smallest daily cost.

Optimization Formulation

$$\min_{L,P,E} \; J(k,L) \tag{1}$$

subject to

$$0 \le P_i(j) \le P_{\max}, \tag{2}$$
$$E_i(j) = E_i(j-1) + P_i(j)\Delta t, \tag{3}$$
$$L(j) = \sum_{m=1}^{N_{\text{EV}}} P_m(j), \quad \forall j \in \{k,\ldots,N\}, \tag{4}$$
$$P_i(j) = 0, \quad \forall j < \text{AT}_i \text{ or } j > \text{DT}_i, \tag{5}$$
$$E_i(\text{DT}_i) = ED_i, \quad \forall i \in \{1,\ldots,N_{\text{EV}}\}, \tag{6}$$
$$J(k,L) = g_{\text{dc}}(k,L) + g_{\text{ec}}(k,L) + g_{\text{oc}}(k,L), \tag{7}$$
$$g_{\text{dc}}(k,L) = r_{\text{nc}}\gamma_{\text{nc}}(k) + r_{\text{op}}\gamma_{\text{op}}(k), \tag{8}$$
$$g_{\text{ec}}(k,L) = \Delta t \sum_{j=k}^{N} r_{\text{ec}}(j) L(j), \tag{9}$$
$$g_{\text{oc}}(k,L) = r_{\text{ec}}^{M} E_m + r_f(g_{\text{dc}}(k,L) + g_{\text{ec}}(k,L)), \tag{10}$$
$$\gamma_{\text{nc}}(k) = \max_{m=k}^{N} L(m), \tag{11}$$
$$\gamma_{\text{op}}(k) = \max_{m \in \mathbb{I}_{\text{op}}(k)} L(m), \tag{12}$$
$$\mathbb{I}_{\text{op}}(k) = \begin{cases} \{N_{\text{start}} \ldots, N_{\text{end}}\}, \text{if } k \le N_{\text{start}} < N_{\text{end}}, \\ \{k,\ldots,N_{\text{end}}\}, \text{if } N_{\text{start}} < k < N_{\text{end}}, \\ \emptyset, \text{otherwise.} \end{cases} \tag{13}$$



NOMENCLATURE

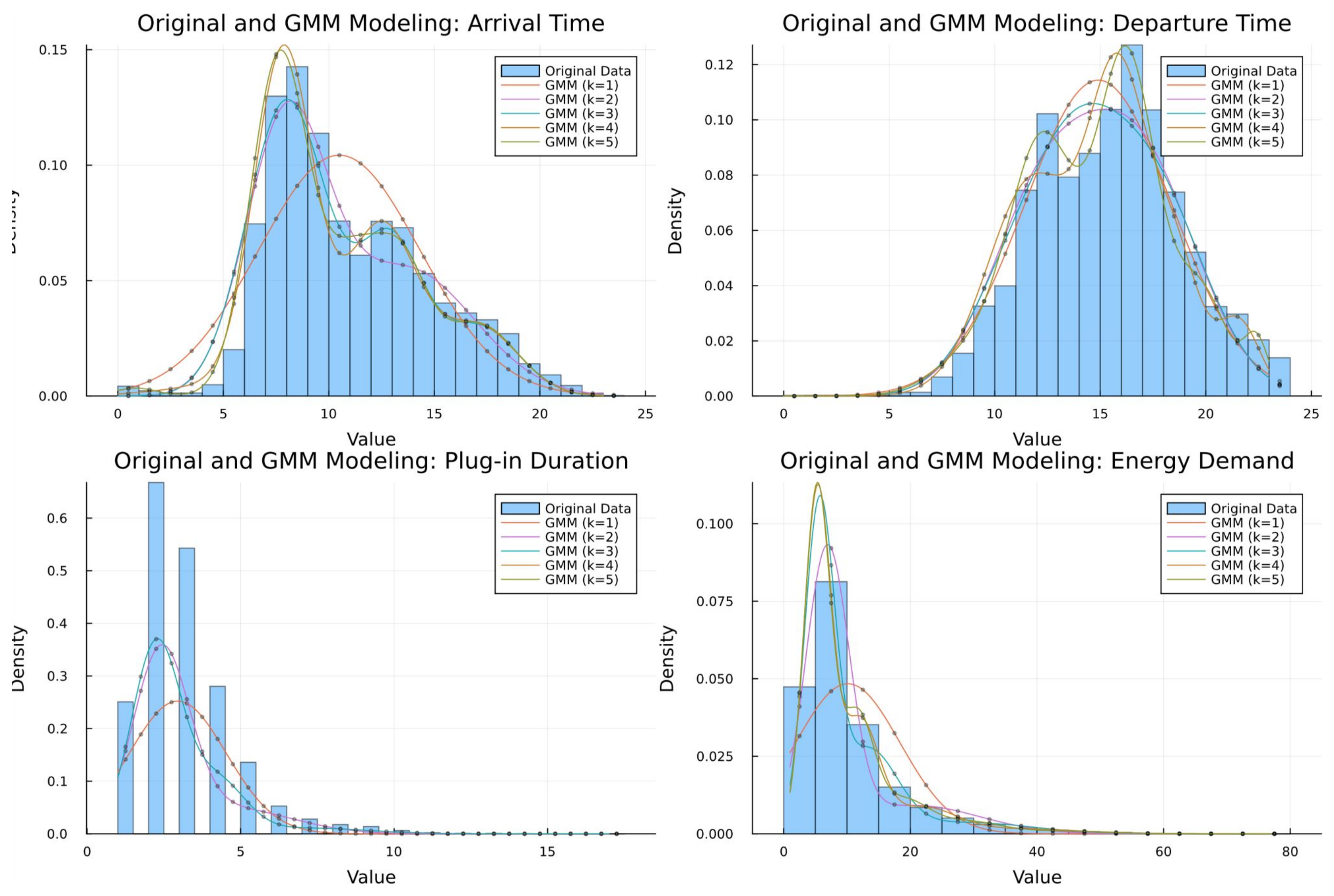
Δt	Time step duration	T	Horizon of MPC
γ	Maximum load	AT	Arrival time
E	Energy dispensed to each EV	DT	Departure time
g	Retail demand and energy cost	DTW	Dynamic Time Warping
J	Cost function	ED	Energy demand
k	Current time step in MPC	EV	Electric Vehicle
L	Net load of charging EVs on the grid	GMM	Gaussian Mixture Model
N	Number of time steps in MPC	kNN	K nearest neighbors
N_{end}	On-peak end time index	L2	Level 2 charging
N_{EV}	Number of EVs	MILP	Mixed-Integer Linear Programming
N_{start}	On-peak start time index	ML	Machine Learning
P	Charging power of each EV	MPC	Model Predictive Control
$P_{\max}$	Maximum charging power of a port	PD	Plug duration
r	Retail rates of demand and energy cost	SOC	State of charge
		TOU	Time of use

PARAMETER VALUES

Symbol	Value	Unit
Δt	0.25	hour
T	24	hour
N	96	–
N_{start}	65	–
N_{end}	84	–
$P_{\max}$	6.6	kW
r_{ec} summer on-peak	0.12628	\$/kWh
r_{ec} summer off-peak	0.10679	\$/kWh
r_{ec} winter on-peak	0.10626	\$/kWh
r_{ec} winter off-peak	0.09506	\$/kWh
r_{op} summer on-peak	28.92	\$/kW
r_{op} winter on-peak	19.23	\$/kW
r_{nc}	24.48	\$/kW

Gaussian Mixture Model of Training Sessions

The distribution of historical EV training sessions are shown in 4 aspects, namely arrival time (AT), departure time (DT), plug-in duration (PD) and ED. Gaussian mixture model (GMM) is used to regress the distribution with expectation maximization.



Cost Analysis

Persistence forecast has the overall minimal cost in executable methods among all days, followed by statistical method. LSTM however does not meet the expectation.